

Security Monitoring with Splunk

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Abstract

There is no gainsaying in the fact that, in today's hybrid cloud environments, organizations, and in fact individuals face increasingly complex cyber threats that often require advanced detection, monitoring, and response mechanisms. Security Information and Event Management (SIEM) tools have however become essential in modern Security Operations Centers (SOCs) for managing diverse log data and detecting malicious activities. This project simulates the role of a Tier 1 SOC Analyst using Splunk as the primary SIEM platform to analyze multi-source logs, detect anomalies, and investigate incidents across a hybrid enterprise environment. Real-world datasets from AWS, Office 365, DNS, Sysmon, endpoint security, and firewall logs were examined. The investigation included user activity monitoring, cloud storage misconfigurations, malware detection, and host anomalies. The outcome highlights Splunk's effectiveness in correlating multi-source logs and supporting incident investigation.

1. Introduction

It is important to note that the rapid adoption of cloud services, coupled with traditional on-premises infrastructure, has introduced new complexities in cybersecurity monitoring. Organizations today face sophisticated threats such as credential theft, misconfigured cloud storage, insider misuse and advanced malware infections, among others. However, to effectively defend against these threats, Security Operations Centers (SOCs) rely on SIEM platforms that provide centralized monitoring, log correlation, and incident investigation.

Splunk has emerged as one of the most powerful SIEM solutions in modern enterprises. It offers a very robust and effective search, visualization, and reporting functionalities. It enables SOC analysts to aggregate, search, and correlate log data from diverse environments, including cloud platforms like AWS and O365, as well as endpoint and network devices.

This capstone project assumes the role of a Tier 1 SOC Analyst (me) tasked with monitoring and investigating suspicious events within a simulated hybrid enterprise environment. Using Splunk Enterprise and the BOTS v3 dataset provided, various security incidents were investigated, ranging from the unauthorized IAM activity to cryptocurrency mining and abnormal endpoint configurations, among others.

The aim of this project is to demonstrate how Splunk can be applied to:

- Detect unauthorized access attempts and misconfigurations.
- Investigate malicious uploads to cloud storage.
- Identify compromised endpoints.
- Correlate logs across multiple sources to build attack storylines.

2. Literature Review

Security Information and Event Management (SIEM) systems have been universally adopted for real-time security monitoring and compliance reporting. Gartner (2021) identifies SIEM tools as critical for detecting advanced threats in hybrid infrastructures. Splunk however, is one of the leading SIEMs that provides indexing, searching, and visualization capabilities that support both Tier 1 monitoring and advanced threat hunting. Recent studies highlight the growing challenge of monitoring multi-cloud environments. AWS misconfigurations, such as publicly accessible S3 buckets, have caused numerous high-profile breaches (Shulman, 2020). In the same vein, endpoint compromise through cryptocurrency mining has increased as attackers monetize computing resources. Splunk's built-in detection logic and custom correlation searches provide a framework for investigating these issues effectively.

3. Methodology

The project was performed in Kali Linux, a controlled virtual lab environment in a bid to avoid accidental exposure of sensitive data.

Environment Setup:

1. First step was the Installation of Splunk Enterprise on a Kali Linux virtual machine.
2. Proceeded to loading of the BOTS v3 dataset into the Splunk instance by unzipping into /opt/splunk/etc/apps.
3. Followed by restarting Splunk to ensure proper indexing of datasets.
4. Data access was thereby confirmed using the search query: index=botsv3 earliest=0.

Tools and Data Sources:

- Splunk Enterprise: Primary SIEM platform.
- BOTS v3 Dataset: Simulated logs from AWS CloudTrail, Office 365, firewall traffic, endpoint logs (Sysmon, antivirus), DNS queries.

Investigation Approach:

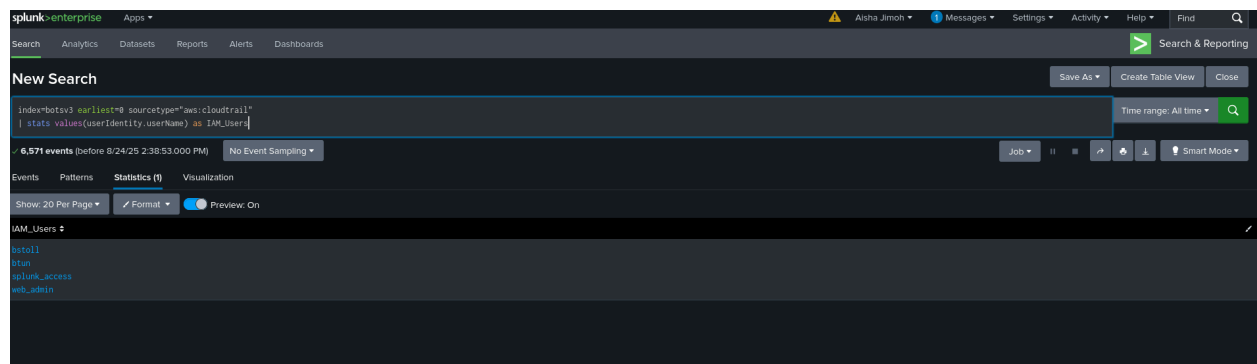
1. Log Analysis: Splunk queries filtered relevant events.
2. Event Correlation: Multi-source data was cross-referenced to identify attack patterns.
3. Incident Identification: Specific deliverables were answered using generated log searches.
4. Documentation: Findings were recorded with Splunk evidence and screenshots.

4. Results

Question 1: IAM Users Accessing AWS Services:

Answer:

- bstoll
- btun
- splunk_access
- web_admin



This result was generated with the search input:

index=botsv3 sourcetype="aws:cloudtrail"

| stats count by userIdentity.userName

Question 2: Processor Number on Web Servers:

Answer: E5-2676 v3.

The screenshot shows the Splunk Search interface. The search bar contains the query: `index=botsv3 sourcetype=hardware "CPU_TYPE" | rex "CPU_TYPE\s+(?<Processor>.+)"`. The search results are displayed in a table format. The table has columns for Time, Event, and VALUE. The results show three events, all with a source of 'hardware' and a sourcetype of 'hardware'. The first event is from 8/20/18 at 3:26:25.000 PM, showing CPU_TYPE as 'Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz'. The second event is from 8/20/18 at 3:24:24.000 PM, showing CPU_TYPE as 'Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz'. The third event is from 8/20/18 at 2:34:50.000 PM, showing CPU_TYPE as 'Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz'.

Time	Event	VALUE
8/20/18 3:26:25.000 PM	CPU_TYPE	Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz
8/20/18 3:24:24.000 PM	CPU_TYPE	Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz
8/20/18 2:34:50.000 PM	CPU_TYPE	Intel(R) Xeon(R) CPU E5-2676 v3 @ 2.40GHz

This result was generated with the search input:

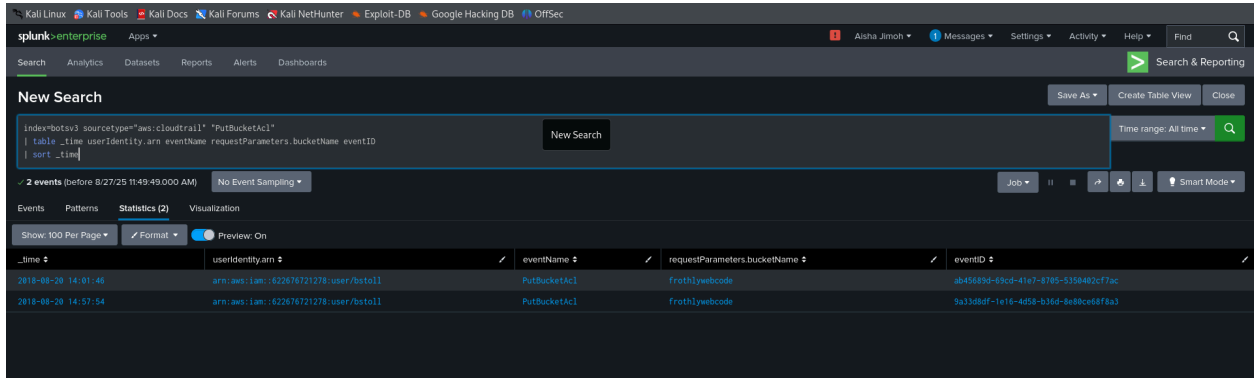
`index=botsv3 sourcetype=hardware "CPU_TYPE"`

`| rex "CPU_TYPE\s+(?<Processor>.+)"`

`| stats values(processor)`

Question 3: Event ID of S3 Bucket Public Access:

Answer: ab45689d-69cd-41e7-8705-5350402cf7ac



The screenshot shows the Splunk Enterprise search interface. The search bar contains the following query:

```
index=botsv3 sourcetype="aws:cloudtrail" "PutBucketAcl"
| table _time userIdentity.arn eventName requestParameters.bucketName eventID
| sort _time
```

The search results show 2 events. The table below displays the data for these events:

_time	userIdentity.arn	eventName	requestParameters.bucketName	eventID
2018-08-20 14:01:46	arn:aws:iam::62287621278:user/bstoll	PutBucketAcl	frothiyeocode	ab45689d-69cd-41e7-8705-5350402cf7ac
2018-08-20 14:57:54	arn:aws:iam::62287621278:user/bstoll	PutBucketAcl	frothiyeocode	ba33d8df-1e16-4d58-b36d-8a80ce6d8a3

This result was generated with the search input:

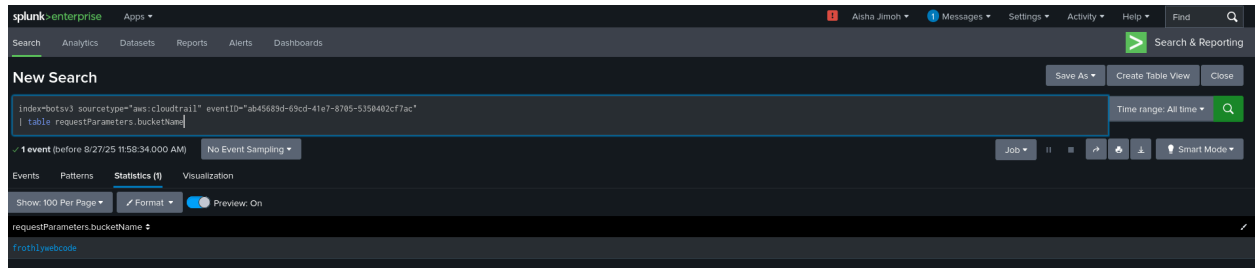
index=botsv3 sourcetype="aws:cloudtrail" "PutBucketAcl"

| table _time userIdentity.arn eventName requestParameters.bucketName eventID

| sort _time

Question 4: Name of the Publicly Accessible S3 Bucket:

Answer: Frothlywebcode.



This result was generated with the search input:

```
index=botsv3 sourcetype="aws:cloudtrail"  
eventID="ab45689d-69cd-41e7-8705-5350402cf7ac"  
  
| table requestParameters.bucketName
```

Answer: OPEN_BUCKET_PLEASE_FIX.txt

splunk-enterprise Apps + **Alina Jimon** Messages Settings Activity Help Find Search & Reporting

Search Analytics Datasets Reports Alerts Dashboards

New Search

Save As Create Table View Close

Index=botsv3 "frothywebcode" ".txt"
| table _time, _raw Time range: All time 🔍

6 events (before 8/27/25 11:59:55:00 AM) No Event Sampling Job + || = ⬆️ ⬇️ ⬅️ Smart Mode +

Events Patterns Statistics (6) Visualization

Show: 100 Per Page Format Preview: On

_time	_raw
2018-08-20 14:02:45	4/018853e74f45b6d45f6dc8c5eff34774548bae481384323cfcdafae1ed FrothyWebcode [20/Aug/2018:13:02:45 +0000] 52.66.146.128 - AB10PC312EC11AC REST.GET.BUCKET - "GET /?prefix=OPEN_BUCKET_PLEASE_FIX.txt&encoding-type=url HTTP/1.1" 200 - 675 - 11 0 "-" BotSv3/1.7.62 Python/2.7.14 Linux/4.14.47-64.38.amzn2.x86_64 BotoCore/1.8.12"
2018-08-20 14:02:44	4/018853e74f45b6d45f6dc8c5eff34774548bae481384323cfcdafae1ed FrothyWebcode [20/Aug/2018:13:02:44 +0000] 52.66.146.128 - DF1ABAD922610BA REST.PUT.OBJECT OPEN_BUCKET_PLEASE_FIX.txt "PUT /OPEN_BUCKET_PLEASE_FIX.txt HTTP/1.1" 200 - - 377 268 0 "-" BotSv3/1.7.62 Python/2.7.14 Linux/4.14.47-64.38.amzn2.x86_64 BotoCore/1.8.12"
2018-08-20 14:33:24	Cloud-init v. 0.7.6 running 'modules/final' at Thu, 20 Jul 2018 00:45:24 +0000 Up 11.83 seconds. % Total % Received % Xferd Average Speed Time Time Current Dload Upload Total Spent Left Speed 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 100 13 100 13 0 0 1731 0 --:--:-- --:--:-- 1500 % Total % Received % Xferd Average Speed Time Time Current Dload Upload Total Spent Left Speed

This result was generated with the search input:

```
index=botsv3 "frothlywebcode" ".txt"
```

	<i>table_time, _raw</i>
--	-------------------------

Answer: 3057116 bits is the first successful one. When converted to MB and rounded up to two decimal places, the size is 2.92 MB.

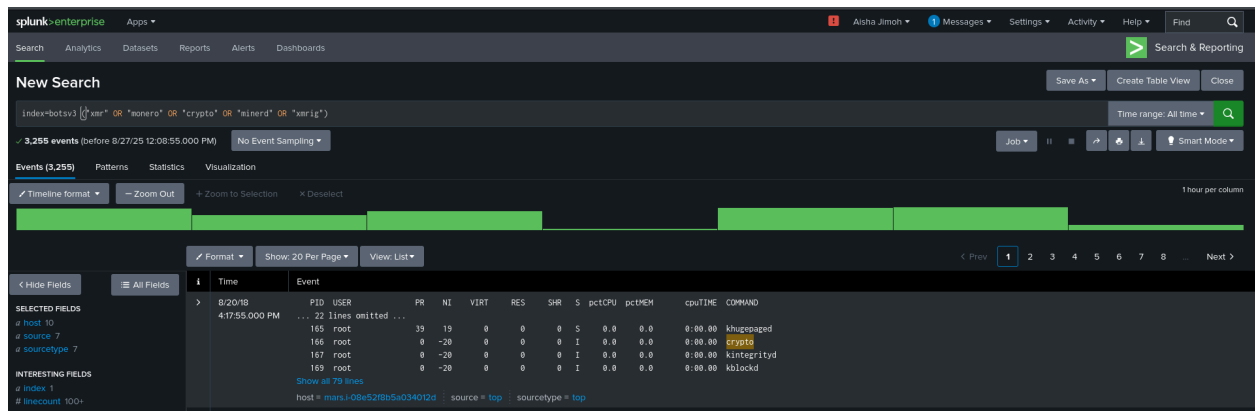


index=botsv3 "REST.PUT.OBJECT" frothlywebcode ".tar.gz"

It is worthy of note that the conversion was done for memory (RAM and cache) so binary was used.

Question 7: Short Hostname of Cryptocurrency Mining Endpoint:

Answer: mars.

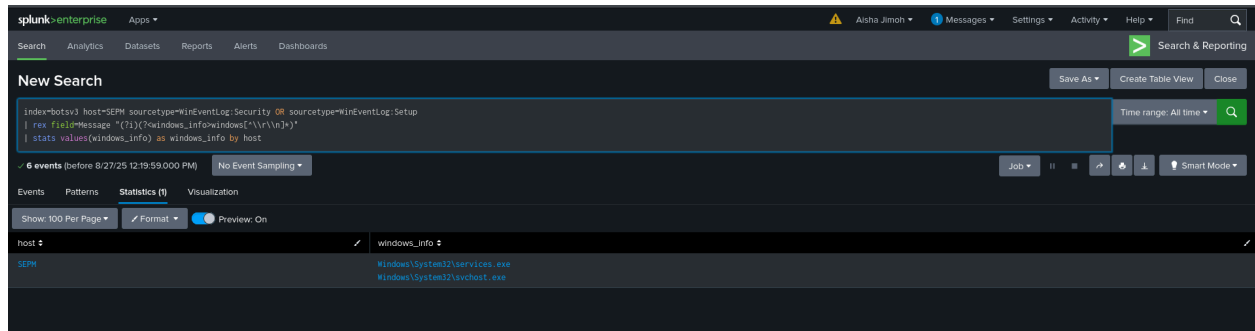


This result was generated with the search input:

index=botsv3 ("xmr" OR "monero" OR "crypto" OR "miner" OR "xmrig")

Question 8: Endpoint Running Different Windows Edition:

Answer: The host is SEPM



This result was generated with the search input:

*index=botsv3 host=SEPM sourcetype=WinEventLog:Security OR
sourcetype=WinEventLog:Setup*

| rex field=Message "(?i)(?<windows_info>windows[^\r\n])"*

| Stats values(windows_info) as windows_info by host

5. Findings

The results to the questions demonstrate Splunk's effectiveness in correlating incidents across multi-source logs. IAM access analysis highlighted potential insider threats, while the S3 misconfiguration reflected a common cloud issue. Malicious uploads revealed the risk of data exfiltration, while the cryptocurrency-mining endpoint illustrated resource abuse. Detecting a different Windows edition among endpoints demonstrated the importance of baseline monitoring. These findings highlight core SOC responsibilities: log monitoring, event correlation, and incident reporting. Deeper forensic investigation would be required for escalation.

6. Conclusion

This project successfully simulated Tier 1 SOC analyst duties in a hybrid cloud environment using Splunk. Multiple incidents were investigated, including IAM misuse, cloud misconfigurations, malicious uploads, mining activity, and endpoint anomalies.

Splunk proved to be an effective SIEM tool, capable of detecting and correlating real-world threats. Future improvements could involve automation with SOAR tools, custom correlation searches, and integration with Kubernetes logs for broader monitoring.

References

1. Let's Defend. (2021). SIEM Hands-On Training.
2. Splunk Inc. (2022). Splunk Enterprise Security Documentation.
3. TryHackMe. (2021). Threat Detection and SIEM Labs.
4. Gartner. (2021). SIEM Market Guide. Gartner Research.
5. Shulman J. (2020). Common AWS Misconfigurations and Security Risks. Cloud Security Alliance.